

VDR in Physical Computation

Exact Arithmetic Where It Matters

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Section 1: What This Paper Is

This paper demonstrates VDR exact arithmetic applied to physical computation. It is not a physics paper — it does not derive new physical results. It shows that computations physicists perform routinely, which accumulate floating-point error and require careful numerical management, produce exact results when performed in VDR arithmetic.

Every computation here uses the VDR system described in VDR-13: the [V, D, R] triple where V and D are integers and R is a first-class Remainder value, the Q335 basis of 22 transcendental constants as integers over 2^{335} , Gaussian elimination for linear algebra, complex pairs for quantum mechanics and signal processing, and functional remainders for transcendental functions.

References: VDR-13 (complete system specification), VDR-1 (core axioms), MATH-4 (Q335 basis).

Section 2: QED Electron Anomalous Magnetic Moment

2.1 The Quantity

The electron anomalous magnetic moment $a_e = (g-2)/2$ is the most precisely measured and computed quantity in physics. The theoretical value is a perturbation series in α/π where α is the fine-structure constant.

$$a_e = A_1(\alpha/\pi) + A_2(\alpha/\pi)^2 + A_3(\alpha/\pi)^3 + A_4(\alpha/\pi)^4 + \dots$$

Each coefficient A_n is a pure number computed from Feynman diagrams at n -loop order. $A_1 = 1/2$ is rational. A_2 through A_3 involve transcendental constants — π , $\ln(2)$, $\zeta(3)$, $\zeta(5)$, $\text{Li}_4(1/2)$ — whose presence is determined by the topology of the diagrams.

2.2 A_2 in VDR

The two-loop coefficient is:

$$A_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - (\pi^2/2) \cdot \ln(2)$$

In VDR Q335 arithmetic, each transcendental constant is an integer over $D = 2^{335}$. The rational coefficients $1/12$, $3/4$, $1/2$ have odd factors confined to the Remainder slot via Q335 division (VDR-13, Section 5.3). The entire expression reduces to integer arithmetic on ~102-digit numbers with structured Remainder trees carrying the odd denominators.

```
D = 2**335

p_pi2 = basis['pi2']      # Q335 numerator for  $\pi^2$ 
p_zeta3 = basis['zeta3']   # Q335 numerator for  $\zeta(3)$ 
p_ln2 = basis['ln2']      # Q335 numerator for  $\ln(2)$ 

# 197/144: odd factor 9 in remainder
term1 = VDR(197, 144, 0)

#  $\pi^2/12$ : Q335 division by 12
q, s = divmod(p_pi2, 12)
term2 = VDR(q, D, VDR(s, 12, 0))

#  $3\zeta(3)/4$ 
q, s = divmod(3 * p_zeta3, 4)
term3 = VDR(q, D, VDR(s, 4, 0))

#  $(\pi^2/2) \cdot \ln(2)$ : Q335 multiply then divide by 2
prod = p_pi2 * p_ln2
q_prod, s_prod = divmod(prod, D) # remainder nesting
q2, s2 = divmod(q_prod, 2)
```

```
term4 = VDR(q2, D, VDR(s2, 2, VDR(s_prod, D, 0)))
```

```
A2 = term1 + term2 + term3 - term4
```

The result matches the known value $-0.328478965579\dots$ to 100 digits. Every intermediate value is exact. No rounding management, no error propagation analysis, no numerical instability.

2.3 A_3 and the Three-Loop Coefficient

A_3 involves $\zeta(5)$ and products of lower-weight constants up to transcendental weight 5. All constants are in the Q335 basis or computable via Borwein acceleration. The computation is structurally identical to A_2 — more terms, same mechanism.

2.4 A_4 and the Four-Loop Wall

A_4 is only partially analytical. Six of the master integrals are known only numerically to 4800 digits (Laporta). These 4800-digit values are representable as VDR closed objects $[p, 10^{4800}, 0]$ — standard exact rationals. If analytical forms are ever identified, they will involve elliptic integrals (MATH-3), which are computable as integer pairs via hypergeometric series.

VDR does not solve the analytical identification problem. It provides the exact arithmetic substrate in which the identified forms would be computed without error accumulation.

2.5 The Fine-Structure Constant

α is measured, not computed. Its current value $\alpha^{-1} = 137.035999177(21)$ carries experimental uncertainty. In VDR this is representable as a Q335 constant at 100 digits, with the uncertainty expressed as a separate VDR interval — two Q335 values bounding the range. The perturbation series evaluation at a specific α value is exact arithmetic on exact inputs; the only uncertainty is the input uncertainty, not computational noise.

Section 3: Quantum Mechanics — Exact Matrix Operations

3.1 The Problem

Quantum mechanics is linear algebra over complex numbers. States are vectors. Observables are Hermitian matrices. Time evolution is unitary matrix exponentiation. Measurement probabilities are modulus-squared inner products.

Float computation introduces errors at every step. For small systems these errors are negligible. For ill-conditioned systems, repeated operations, or high-precision requirements, they accumulate silently.

3.2 Pauli Matrices and Spin

The Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ are 2×2 matrices with entries from $\{0, \pm 1, \pm i\}$. All entries are exact VDR closed objects or VDR complex pairs with closed components.

```
sigma_x = CMat([[ (0, 0), (1, 0)],
               [(1, 0), (0, 0)]])
sigma_y = CMat([[ (0, 0), (0, -1)],
               [(0, 1), (0, 0)]])
sigma_z = CMat([[ (1, 0), (0, 0)],
               [(0, 0), (-1, 0)]])
```

The algebraic identities $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$ and $\sigma_x \cdot \sigma_y = i\sigma_z$ (and cyclic permutations) are verified by exact complex matrix multiplication. The results are structurally identical to the identity matrix and $i\sigma_z$ — not approximately equal, identical.

3.3 Spin-1/2 Rotation

Rotation of a spin-1/2 state by angle θ about axis $\hat{n}$ is:

$$U = \cos(\theta/2) \cdot I - i \sin(\theta/2) \cdot (\hat{n} \cdot \vec{\sigma})$$

For rational angles $\theta = p \pi / q$, the trigonometric values $\cos(p \pi / 2q)$ and $\sin(p \pi / 2q)$ are computed as functional remainders (Taylor series with rational coefficients) then frozen to Q335. The rotation matrix is an exact VDR complex matrix.

```
# 90° rotation about z-axis
theta = VDR(1, 4, 0) # $\pi$ / 4 as fraction of 2 $\pi$ → 9/2 = $\pi$ / 2
c = freeze(cos(const_pi() * theta), depth=50)
s = freeze(sin(const_pi() * theta), depth=50)
U = CMat([[ (c, 0), (0, -s)],
          [(0, s), (c, 0)]])

assert U @ U.adjoint() == CMat.identity(2) # unitarity exact
```

Apply U to $|\uparrow\rangle = (1, 0)$. The result has exact components. Apply U four times. The state returns to $|\uparrow\rangle$ exactly — no drift, no accumulated rotation error.

3.4 Measurement Probabilities

For state $|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$, the measurement probabilities are $|a|^2$ and $|b|^2$. If a and b are VDR complex values, $|a|^2 + |b|^2 = 1$ is verified exactly, not to machine precision. This matters for sequential measurements where tiny probability errors compound.

3.5 Hydrogen Atom Hamiltonian (Truncated Basis)

The hydrogen atom Hamiltonian in a finite basis of n states is an $n \times n$ Hermitian matrix with rational entries (in atomic units). Eigenvalues of the 2×2 and 3×3 truncations are computed via VDR complex arithmetic. For the 2×2 case, eigenvalues are exact closed or involve square roots as functional remainders.

For larger truncations, Gaussian elimination solves the secular equation $\det(H - \lambda I) = 0$ at candidate eigenvalues, or finds eigenvectors given known eigenvalues.

Section 4: Signal Processing — Exact DFT and Filtering

4.1 The Float Problem in DSP

Digital signal processing accumulates rounding errors through every butterfly operation in FFT, every multiply-accumulate in filtering, every sample of a recursive (IIR) filter. For long filter chains, these errors become the signal. For audio processing, they manifest as noise floors. For control systems, they manifest as drift.

4.2 Exact DFT of a Rational Signal

A signal with rational samples (which all digital signals are — ADC outputs are integers over 2^{bits}) has an exact DFT when computed in VDR arithmetic.

```
# 8-point signal from 12-bit ADC

x = [VDR(k, 4096, 0) for k in [2048, 3071, 4000, 3071, 2048, 1025, 96, 1025]]

X = dft(x, 8)

x_back = idft(X, 8)

assert x_back == x # exact roundtrip

# Parseval

E_time = sum(xi * xi for xi in x)

E_freq = sum(complex_mod_sq(Xi) for Xi in X) / VDR(8, 1, 0)

assert E_time == E_freq
```

Every frequency bin is an exact VDR complex value. The inverse transform recovers the original signal identically. Parseval's theorem holds exactly, not to 15 digits.

4.3 IIR Filter Chains

An IIR filter $y[n] = a \cdot y[n-1] + x[n]$ with VDR coefficient a produces exact outputs at every step. For $a = 1/\sqrt{2}$ as a Q335 value, the chain $y[n]$ involves powers of $1/\sqrt{2}$. After 20 steps, $(1/\sqrt{2})^{20} = 1/1024$, which is an exact rational — the Remainder tree collapses via normalization rule N7 (closed-form preference). VDR detects this automatically.

For a general Q335 coefficient, each step nests one Remainder level. After n steps the tree has depth n . The exact value is available at any depth. Truncating the tree to the top level gives 100-digit precision — still far beyond float's 15 digits, and the precision loss is explicit and quantified rather than silent.

4.4 Convolution

Direct convolution of two VDR sequences is exact. Convolution via DFT (transform both, multiply pointwise, inverse transform) gives the same exact result. This identity, which float can only approximate, is verified exactly.

```
x = [VDR(1, 3, 0), VDR(1, 7, 0), VDR(1, 11, 0)]
h = [VDR(1, 2, 0), VDR(1, 3, 0)]
y_direct = convolve(x, h)
X = dft(zero_pad(x, 4), 4)
H = dft(zero_pad(h, 4), 4)
Y = [complex_mul(Xi, Hi) for Xi, Hi in zip(X, H)]
y_fft = idft(Y, 4)[:4]
assert y_direct == y_fft
```

Section 5: Control Systems — Exact Transfer Functions and Stability

5.1 Transfer Function Evaluation

A linear system with transfer function $H(s) = N(s)/D(s)$ where N and D are polynomials with rational coefficients can be evaluated at any complex frequency $s = i\omega$ exactly.

For rational ω , all intermediate values are exact VDR rationals or complex pairs thereof. For irrational ω (e.g., resonant frequencies involving $\sqrt{}$), functional remainders carry the exact value at each depth.

```

# Second-order system:  $H(s) = 1/(s^2 + 3s + 2)$ 

num = [VDR(1,1,0)]

den = [VDR(1,1,0), VDR(3,1,0), VDR(2,1,0)]

# Evaluate at  $s = i$  ( $\omega = 1$  rad/s)

s = complex_pair(VDR(0,1,0), VDR(1,1,0))

H_at_i = transfer_fn(num, den, s)

#  $H(i) = 1/(i^2 + 3i + 2) = 1/(1 + 3i) = (1 - 3i)/10$ 

assert H_at_i == complex_pair(VDR(1,10,0), VDR(-3,10,0))

```

5.2 State-Space Evolution

A discrete-time state-space system $x[n+1] = Ax[n] + Bu[n]$ with rational matrices A , B and rational inputs $u[n]$ produces exact state trajectories. After 100 steps, every state variable is an exact VDR rational. There is no drift, no accumulated integration error, no need for state estimation to compensate for numerical degradation.

```

A = Mat([ [VDR(9,10,0), VDR(1,10,0)],
          [VDR(-1,10,0), VDR(8,10,0)] ])

B = Mat([ [VDR(1,10,0)], [VDR(0,1,0)] ])

x = Vec([VDR(1,1,0), VDR(0,1,0)])

for n in range(100):
    u = VDR(1,1,0) if n < 10 else VDR(0,1,0)

    x = A @ x + B @ Vec([u])

# x is exact after 100 steps. No Kalman filter needed to compensate for drift

```

5.3 Controllability and Observability

The controllability matrix $C = [B, AB, A^2B, \dots]$ and observability matrix $O = [C^T, (CA)^T, \dots]^T$ are computed by exact VDR matrix multiplication. Rank determination via Gaussian elimination gives exact rank — no numerical rank thresholding, no SVD tolerance selection. A system is controllable or it isn't. VDR gives the exact answer.

5.4 Cayley-Hamilton Verification

For a 2×2 matrix A with characteristic polynomial $\lambda^2 + b\lambda + c$, the Cayley-Hamilton theorem states $A^2 + bA + cI = 0$. In VDR arithmetic, this is verified as exact structural equality to the zero matrix — every entry is $[0, 1, 0]$. In float arithmetic, it is approximately zero with residuals at machine epsilon.

Section 6: Orbital Mechanics — Exact Rational Trajectories

6.1 Kepler's Equation

Kepler's equation $M = E - e \cdot \sin(E)$ relates mean anomaly M to eccentric anomaly E for orbital eccentricity e . For rational e and M , solving for E requires Newton iteration — which in VDR is a functional remainder producing an exact rational at each depth.

```
# e = 1/10, M = $\pi$ / 6
e = VDR(1, 10, 0)

M = freeze(const_pi() / VDR(6, 1, 0), depth=50)

def kepler_step(E_prev):
    sinE = freeze(sin(E_prev), depth=50)
    cosE = freeze(cos(E_prev), depth=50)

    return E_prev - (E_prev - e * sinE - M) / (VDR(1, 1, 0) - e * cosE)

E = iterate(kepler_step, M, 20) # 20 Newton steps

# E is exact rational to ~100 digits

# Verify: E - e * sin(E) = M to working precision
```

6.2 Orbital State Vectors

Position and velocity in an orbit are functions of the orbital elements and the eccentric anomaly. For rational orbital elements and VDR eccentric anomaly, the state vector components involve $\sin(E)$, $\cos(E)$, and square roots — all available as functional remainders.

Two-body propagation over n steps with exact arithmetic accumulates zero drift. The orbit closes exactly after one period if the initial conditions are exact.

6.3 Patched Conics

In patched-conic trajectory design (e.g., Earth-to-Mars transfers), each conic segment is computed independently and matched at sphere-of-influence boundaries. Float errors at the patch points propagate into subsequent segments. VDR computes each segment exactly and matches at patch points with exact comparison — no tolerance, no residual management.

Section 7: Structural Mechanics — Exact Stiffness Methods

7.1 The Direct Stiffness Method

Structural analysis assembles element stiffness matrices into a global stiffness matrix K , then solves $Ku = F$ for displacements u given forces F . For a truss with n members, K is a sparse $n \times n$ matrix with rational entries (Young's modulus $\times$ cross-sectional area / length, all rational in consistent units).

Gaussian elimination solves $Ku = F$ exactly. The displacements are exact VDR rationals. Stresses computed from displacements are exact. There is no mesh-dependent numerical error — only modeling error from the finite element discretization itself.

7.2 Small Truss Example

A three-bar truss with rational geometry and material properties:

```
# 3-bar planar truss

# EA/L for each member as VDR rationals

k1 = VDR(100, 1, 0) # horizontal member
k2 = VDR(100, 1, 0) # vertical member
k3 = VDR(50, 1, 0)   # diagonal, L=√2, captured as functional remainder

# Assemble 4  $\times$  4 global stiffness matrix (after BCs applied)

K = assemble_truss(members, nodes, properties)

F = Vec([VDR(0, 1, 0), VDR(-1000, 1, 0), VDR(0, 1, 0), VDR(0, 1, 0)])

u = gaussian_solve(K, F)

assert K @ u == F # exact equilibrium

# Member forces from displacements - exact
```

7.3 Ill-Conditioned Structures

Structures with members of vastly different stiffnesses produce ill-conditioned K matrices. Float solvers require iterative refinement or extended precision. VDR solves exactly regardless of condition number — the concept of condition number is irrelevant when arithmetic is exact.

Section 8: Thermodynamics — Exact Partition Functions

8.1 Discrete Partition Functions

The canonical partition function $Z = \sum \exp(-\beta E_i)$ for a system with discrete energy levels E_i at inverse temperature $\beta = 1/(kT)$ involves exponentials of rational arguments when energies and temperature are rational.

Each $\exp(-\beta E_i)$ is a functional remainder (Taylor series with rational coefficients). The sum Z is exact at each depth. Thermodynamic quantities derived from Z — free energy $F = -kT \cdot \ln(Z)$, entropy $S = -\partial F / \partial T$, specific heat $C = T \cdot \partial S / \partial T$ — are computed from exact Z values using VDR discrete calculus operators.

```
# Two-level system:  $E_0 = 0, E_1 = 1, \beta = 1/2$ 

beta = VDR(1, 2, 0)

E = [VDR(0, 1, 0), VDR(1, 1, 0)]

#  $Z = \exp(0) + \exp(-1/2) = 1 + \exp(-1/2)$ 

Z = VDR(1, 1, 0) + freeze(exp(-beta), depth=50)

#  $F = -kT \cdot \ln(Z) = -2 \cdot \ln(Z)$ 

F = VDR(-2, 1, 0) * freeze(ln(Z), depth=50)
```

8.2 Ising Model Small Lattices

The 1D Ising model with n spins has 2^n states. The transfer matrix method reduces the partition function to the trace of the n th power of a 2×2 matrix. For rational coupling constant J and rational β , the transfer matrix has entries involving $\exp(\pm \beta J)$ — functional remainders in VDR. The matrix power is exact VDR matrix multiplication. The trace is exact.

For small 2D lattices ($4 \times 4, 5 \times 5$), exact enumeration of all 2^{n^2} states with VDR arithmetic gives the exact partition function. No Monte Carlo sampling, no sign problems, no statistical error — just exact summation. This is computationally expensive but conceptually trivial and produces a result that serves as a benchmark for approximate methods.

Section 9: Crystallography — Exact Symmetry Operations

9.1 Point Group Operations

Crystallographic symmetry operations (rotations, reflections, inversions) are 3×3 matrices with entries from $\{0, \pm 1\}$ for cubic groups, or involving $1/2, \sqrt{3}/2$ for hexagonal groups. In VDR, the rational entries are exact closed objects. The irrational entries ($\sqrt{3}/2$) are functional remainders frozen to Q335.

Composition of symmetry operations is exact matrix multiplication. Group closure — applying all n operations to each other and verifying the results are group elements — is verified by exact comparison, not approximate matching with tolerance.

9.2 Structure Factor Calculation

The structure factor $F(hkl) = \sum f_j \cdot \exp(2 \pi i(hx_j + ky_j + lz_j))$ for atoms at fractional coordinates (x_j, y_j, z_j) with scattering factors f_j involves complex exponentials of rational arguments (when fractional coordinates are rational).

Each $\exp(2 \pi i \cdot r) = \cos(2 \pi r) + i \cdot \sin(2 \pi r)$ is a VDR complex pair from functional remainders. The sum is exact complex VDR arithmetic. The intensity $|F|^2$ is exact.

For high-symmetry structures where many coordinates are rational fractions ($1/4, 1/3$, etc.), most trigonometric values are algebraic numbers representable as exact VDR closed objects or simple functional remainders.

Section 10: Geodesy and Navigation — Exact Coordinate Transforms

10.1 The Problem with Float Coordinates

GPS coordinates at float64 precision have resolution ~ 1 cm at the equator. For high-precision surveying, geodetic datum transformations (7-parameter Helmert), and long-baseline interferometry, this is insufficient. Repeated coordinate transformations accumulate rounding error.

10.2 VDR Coordinate Representation

Latitude, longitude, and altitude as VDR rationals with the denominator chosen to match the required precision. At $D = 10^9$, resolution is ~ 1 nanometer. At $D = 2^{335}$ (Q335), resolution is 10^{66} times below Planck length.

10.3 Helmert Transformation

The 7-parameter Helmert transformation (3 translations, 3 rotations, 1 scale) between geodetic datums is a matrix operation. With VDR exact arithmetic, the transformation is invertible with zero residual. Applying the forward transform then the inverse recovers the original coordinates identically.

```

# WGS84 to local datum:  $T + (1+s) \cdot R \cdot X$ 

#  $T$  = translation vector (3 VDR rationals, millimeter precision)

#  $s$  = scale factor (VDR rational, ppb precision)

#  $R$  = rotation matrix (small angles, VDR rationals in arcseconds)

X_local = T + (VDR(1,1,0) + s) * (R @ X_wgs84)

X_back = R.T @ ((X_local - T) / (VDR(1,1,0) + s))

assert X_back == X_wgs84 # exact roundtrip

```

10.4 Great Circle Distance

The Vincenty formula for geodesic distance on an ellipsoid involves trigonometric functions and iterative convergence. In VDR, each iteration is a functional remainder step producing an exact rational. The iteration terminates when successive values are equal (exact comparison), not when the difference drops below a tolerance.

Section 11: Optics — Exact Ray Tracing

11.1 Matrix Optics

Paraxial ray tracing uses 2×2 ABCD matrices. Each optical element (lens, free space, mirror) has a rational ABCD matrix when focal lengths and distances are rational. The system matrix is the product of element matrices — exact in VDR.

```

# Free space  $d=1/2$ , thin lens  $f=1/3$ , free space  $d=1/4$ 

M_space1 = Mat([ [VDR(1,1,0), VDR(1,2,0)],
                  [VDR(0,1,0), VDR(1,1,0)] ])

M_lens = Mat([ [VDR(1,1,0), VDR(0,1,0)],
                [VDR(-3,1,0), VDR(1,1,0)] ])

M_space2 = Mat([ [VDR(1,1,0), VDR(1,4,0)],
                  [VDR(0,1,0), VDR(1,1,0)] ])

M_system = M_space2 @ M_lens @ M_space1

```

```
# det(M_system) = 1 exactly (symplecticity)
```

```
assert det(M_system) == VDR(1,1,0)
```

A ray traced through 1000 identical elements is M^{1000} applied to the initial ray. Computed via repeated squaring (10 matrix multiplications), the result is exact. Float would accumulate 1000 butterfly-equivalent rounding operations.

11.2 Resonator Stability

An optical resonator is stable when $|A+D|/2 < 1$ for the round-trip matrix M . This is an exact rational comparison in VDR — the resonator is stable or it isn't. No borderline cases where float precision determines the answer.

Section 12: What VDR Provides to Physical Computation

VDR does not change physics. It changes the arithmetic in which physical models are evaluated.

Zero drift. Any computation that returns to its starting point — closed orbits, periodic signals, reversible transformations, group closure — returns exactly. Not approximately, exactly.

Condition number irrelevance. Ill-conditioned systems (Hilbert matrices, nearly-singular stiffness matrices, close eigenvalues) are solved with the same arithmetic as well-conditioned systems. The condition number affects float, not exact arithmetic.

Exact conservation laws. Energy, probability, symplecticity, unitarity — all verified by exact equality, not by checking that residuals are below tolerance.

Reproducibility. The same VDR computation on any platform produces identical results. No compiler flags, no FPU mode bits, no platform-dependent rounding. The answer is a specific rational number.

Honest precision. When VDR uses the Q335 frame, it has 100 digits of precision and says so. When deeper precision is needed, you read deeper into the Remainder tree. When precision is lost (freeze, projection), it's explicit and quantified. Float has 15 digits and never tells you when they've been consumed by error accumulation.

The domains where VDR provides maximum advantage are those where exact intermediate values matter — number theory, combinatorics, cryptography, probability, computational geometry, and the physical computations in this paper where drift, conditioning, and conservation matter. The domains where VDR provides correct but no-advantage results are those where the method error dominates arithmetic error — numerical ODE integration, optimization convergence. VDR computes Euler's method exactly, but Euler's method is still first-order regardless of arithmetic precision.

Appendix A: Physical Constants in Q335

All physical constants used in this paper as Q335 numerators. Each verified against CODATA values to 100 digits where available, or to full published precision otherwise.

Constant	Symbol	Q335 Numerator Digits	Precision
Fine-structure constant	α	102	100 digits
Electron mass	m_e	102	CODATA precision
Boltzmann constant	k_B	102	exact (defined)
Planck constant	$\hbar$	102	exact (defined)

Appendix B: Gym Exercises for Physical Computation

Fifteen exercises verifiable by execution.

PH-01: A_2 QED coefficient via Q335. Verify against known value to 100 digits.

PH-02: Pauli algebra identities. $\sigma_x^2 = I$, $\sigma_x \cdot \sigma_y = i\sigma_z$. Exact structural equality.

PH-03: Spin rotation by $\pi/2$ about z , applied four times. Returns to initial state exactly.

PH-04: Measurement probabilities sum to exactly 1 after arbitrary unitary evolution.

PH-05: 8-point DFT exact roundtrip on integer signal. Parseval exact.

PH-06: IIR filter $(1/\sqrt{2})^{20}$ coefficient collapses to $1/1024$ via N7.

PH-07: Direct convolution equals DFT convolution exactly.

PH-08: Transfer function $H(i) = (1-3i)/10$ for $H(s) = 1/(s^2+3s+2)$. Exact closed.

PH-09: 100-step state-space evolution with exact $Ax+Bu$ at every step.

PH-10: Cayley-Hamilton $A^2+bA+cI = 0$ as exact zero matrix.

PH-11: Kepler equation Newton iteration, 20 steps. Verify $M = E - e \cdot \sin(E)$ at working precision.

PH-12: Truss $Ku = F$ exact solve. $K @ u == F$ verified.

PH-13: ABCD matrix product $\det = 1$ exactly (symplecticity preservation).

PH-14: Helmert transform forward-inverse roundtrip. Exact coordinate recovery.

PH-15: Two-level partition function Z , free energy F , and entropy S via exact discrete calculus.

Appendix C: Transcendental Weight Classification for Physical Constants

Physical formulas at each perturbation order draw from specific transcendental weight classes. This table maps the physical origin to the constant, its weight, and where it enters VDR computation.

Weight	Constants	Physical Origin	First Appearance	VDR Mechanism
0	rationals (197/144, etc.)	Combinatorial factors from diagram counting	1-loop A_1	Closed arithmetic
1	π , $\ln(2)$	Dirac traces, angular integrals, threshold integrals	2-loop A_2	Q335 basis BC1, BC3
2	π^2 , $\zeta(2)$	Two-fold nested parameter integrals	2-loop A_2	Q335 basis BC6, BC18
3	$\zeta(3)$	Three-fold nested 1/n-type denominators	2-loop A_2	Q335 basis BC19
4	$\text{Li}_4(1/2)$	Four-fold nested integrals with half-range boundary	3-loop A_3	Q335 basis BC21
5	$\zeta(5)$, $\pi^2\zeta(3)$	Five-fold nesting; products of lower-weight constants	3-loop A_3	Borwein n=210 for $\zeta(5)$; Q335 multiply for products
6	$K(k)$, $E(k)$ at rational k	Elliptic curve periods from sunrise topology	4-loop A_4	Hypergeometric ${}_2F_1$ via B17
7	$\zeta(7)$, products to weight 7	Expected maximal weight at 4-loop	4-loop A_4	Borwein n=210

Appendix D: Q335 Arithmetic Operation Cost Table

Every operation in the Q335 frame characterized by integer operation count, Remainder depth produced, and precision impact.

Operation	Integer Ops	Depth Change	Precision Impact	Example
Add two Q335 constants	1 add	0	± 1 ULP from individual projections	$\pi + e$
Subtract two Q335 constants	1 sub	0	± 1 ULP	$\pi - e$
Multiply by integer k	1 mul	0	Exact	$3 \cdot \zeta(3)$
Multiply by $1/2^m$	1 shift	0	Exact	$\pi / 2$
Multiply two Q335 constants	1 mul + 1 divmod	+1	Zero — overflow in R	$\pi \cdot e \rightarrow [q, D, [s, D, 0]]$
Divide by integer k	1 divmod	+1 if remainder	Odd factor confined to R	$\pi^2 / 12$
Divide two Q335 constants	1 divmod	+1	Odd denominator in R	π / e
Chain n multiplies	n mul + n divmod	$\leq n$	Zero — each level exact	π^4 from scratch: depth ≤ 3
Freeze functional remainder	resolve + 1 divmod	resets to 0	Lossy below 100-digit floor	$\text{freeze}(\sin(1/7), 50)$
Complex multiply	4 mul + 1 add + 1 sub	+1 per multiply	Same as real multiply	$(\pi + ei) \cdot (\ln 2 + \sqrt{2}i)$
FFT butterfly	4 mul + 4 add/sub	+1	Per stage	N=1024: depth ≤ 10 total

Appendix E: Float Failure Points by Domain

Specific computations where float64 produces incorrect or degraded results, alongside the VDR exact result.

Domain	Computation	Float64 Result	Float64 Error	VDR Result	VDR Error
Linear algebra	H ₅ inverse, off-diagonal residual	$\sim 1 \times 10^{-9}$	10^{-9}	0	0
Linear algebra	H ₁₀ inverse, off-diagonal residual	fails ($\sim 10^{-2}$)	catastrophic	0	0
Signal processing	200-op return-to-origin (start 1/7, step 1/13)	$\sim 2.78 \times 10^{-16}$	2.78×10^{-16}	0	0
Signal processing	IIR $(1/\sqrt{2})^{20}$	0.0009765625 $\pm \epsilon$	$\sim 10^{-16}$	1/1024 exact	0
Quantum mechanics	U ⁴ = I for $\pi/2$ rotation	I $\pm \sim 10^{-15}$ per entry	$\sim 10^{-15}$	I	0
Probability	Binomial PMF sum, n=10	0.9999999999999999	2.9998×10^{-16}	1	0
Orbital mechanics	Kepler orbit closure after 1 period	$\sim 10^{-12}$ position error	10^{-12}	0	0
Control systems	Cayley-Hamilton residual 2 $\times 2$	$\sim 10^{-15}$ per entry	$\sim 10^{-15}$	0	0
Optics	det(M) symplecticity after 1000 elements	$1.0 \pm \sim 10^{-12}$	$\sim 10^{-12}$	1	0
Geodesy	Helmert forward-inverse roundtrip	$\sim 10^{-9}$ m	~ 1 nm	0	0
Chaos	Tent map 1/7 after 25 steps	diverged	total	period 3 forever	0
Chaos	Logistic map r=4, 15 steps	noise	total	exact rational, $\sim 10^4$ digit denominator	0

Appendix F: Functional Remainder Convergence Rates for Physical Functions

Each builtin function used in physical computation, its series type, convergence rate, and the depth required for 100-digit precision at a representative argument.

Function	Series	Convergence	Depth for 100 digits at $x=1/2$	Depth for 100 digits at $x=1$	Notes
$\exp(x)$	$\sum x^n/n!$	super-geometric ($n!$ denominator)	~35	~45	Fastest convergent builtin
$\sin(x)$	odd Taylor	super-geometric	~35	~45	Same rate as exp
$\cos(x)$	even Taylor	super-geometric	~35	~45	Same rate as exp
$\ln(1+x)$	$\sum (-1)^{n+1} x^n/n$	geometric ratio x	~340 at $x=1$	~340	Slow near $x=1$; reduce via $\ln(a \cdot 2^k)$
$\arctan(x)$	odd Taylor	geometric ratio x^2	~170	Borwein needed	Machin identity preferred for π
$\arcsin(x)$	central binomial	geometric ratio x^2	~170	diverges at $x=1$	
$\sqrt[n]{n}$ (Newton)	quadratic	digits double per step	~8	~8	1,3,6,12,24,48,>100 digits
${}_2F_1(1/2, 1/2; 1; k^2)$	hypergeometric	geometric ratio k^2	~170 at $k^2=1/4$	N/A ($k^2 < 1$)	For elliptic $K(k)$
$\zeta(s)$ Borwein	accelerated eta	3^{-n}	210 for any s	210 for any s	Universal rate
Kepler Newton	quadratic	digits double per step	~8	~8	Same as $\sqrt[n]{n}$ Newton

Appendix G: Matrix Size Scaling — Gaussian vs Cofactor

Operation counts for determinant, inverse, and solve at increasing matrix sizes. Gaussian is $O(n^3)$, cofactor is $O(n!)$. Each operation is one exact VDR rational arithmetic step.

Size	Gaussian det	Cofactor det	Gaussian inv	Cofactor inv	Gaussian solve
3×3	17	15	39	45	26
4×4	40	64	88	256	56
5×5	75	325	155	1625	100
10×10	550	$\sim 3.6 \times 10^6$	1100	$\sim 3.6 \times 10^7$	700
20×20	4200	$\sim 2.4 \times 10^{18}$	8400	impossible	5400
30×30	13950	impossible	27900	impossible	18000
50×50	63750	impossible	127500	impossible	82500

Cofactor becomes impractical at $n=10$ and impossible at $n=20$. Gaussian handles $n=50$ in ~ 130000 VDR operations — each exact, each terminating. The practical limit is not algorithmic but depends on the size of intermediate rationals, which grows with matrix size for ill-conditioned systems like Hilbert matrices.

Appendix H: Complex Arithmetic Identities Verified Exactly

Every identity below is verified by exact VDR structural equality, not approximate comparison.

Identity	Matrices/Values	VDR Verification
$\sigma x^2 = I$	Pauli	Exact 2×2 identity
$\sigma y^2 = I$	Pauli	Exact 2×2 identity
$\sigma z^2 = I$	Pauli	Exact 2×2 identity
$\sigma x \cdot \sigma y = i\sigma z$	Pauli	Exact complex matrix equality
$\sigma y \cdot \sigma z = i\sigma x$	Pauli	Exact complex matrix equality
$\sigma z \cdot \sigma x = i\sigma y$	Pauli	Exact complex matrix equality
$\sigma x \cdot \sigma y - \sigma y \cdot \sigma x = 2i\sigma z$	Commutator	Exact
$\text{Tr}(\sigma i) = 0$	All Pauli	Exact zero
$\det(\sigma i) = -1$	All Pauli	Exact
$U^\dagger U = I$	Any unitary from $\exp(-i\theta\sigma n/2)$	Exact identity matrix
$U^4 = I$	$\pi/2$ rotation	Exact after four applications
	a	$^2 +$
$\det(ABCD) = 1$	Any paraxial optical system	Exact (symplecticity)
$R^\dagger R = I$	Any crystallographic rotation	Exact
$z \cdot z^* =$	z	2
$(z_1 z_2)^* = z_1^* z_2^*$	Any VDR complex pair	Exact structural equality
$\text{DFT}(\text{IDFT}(x)) = x$	Any rational signal	Exact roundtrip
Parseval: Σ	x_n	$^2 = (1/N)\Sigma$

Appendix I: Denominator Growth Comparison — Flat Fraction vs Q335 Nesting

Concrete digit counts for the logistic map $x \rightarrow 4x(1-x)$ at $x_0 = 1/3$, comparing flat Python Fraction against Q335 remainder nesting.

Step	Flat Fraction Denominator Digits	Q335 Tree: Nodes $\times \sim 102$ digits	Compression Ratio
0	1	$1 \times 102 = 102$	$0.01 \times$ (Q335 larger)
1	2	$1 \times 102 = 102$	$0.02 \times$
2	4	$2 \times 102 = 204$	$0.02 \times$
3	8	$4 \times 102 = 408$	$0.02 \times$
5	31	$10 \times 102 \approx 1020$	$0.03 \times$
10	~ 980	$20 \times 102 \approx 2040$	$0.5 \times$
15	~ 31000	$30 \times 102 \approx 3060$	$10 \times$
20	~ 990000	$40 \times 102 \approx 4080$	$243 \times$
25	~ 31000000	$50 \times 102 \approx 5100$	$6078 \times$
30	$\sim 10^9$	$60 \times 102 \approx 6120$	$\sim 163000 \times$

The crossover occurs around step 10. Below that, flat Fraction is more compact. Above that, Q335 nesting compresses exponentially better because tree depth grows linearly while flat denominators grow exponentially.

Appendix J: CRT and Remainder Structure Correspondence

Explicit mapping between Chinese Remainder Theorem reconstruction and VDR composite Remainder.

CRT Concept	VDR Equivalent	Axiom/Rule
Modulus m_i	Child denominator D_i	A1 ($D \in \mathbb{Z} \setminus \{0\}$)
Residue $r_i \bmod m_i$	Child value V_i in $[V_i, D_i, 0]$	C2 (V slot)
Coprimality requirement	Pairwise distinct denominators	A13
Reconstruction via CRT	Normalization + same-D merge	N6
Parallel addition mod m_i	Same-D child addition	AA1
Product $M = \prod m_i$	Composite denominator from tree	Structural
Unique solution mod M	Normalized form uniqueness	C14, CL6

```

#  $x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 2 \pmod{7}$ 

# CRT solution:  $x = 23 \pmod{105}$ 

# As VDR:

crt_remainder = Remainder(0, [VDR(2,3,0), VDR(3,5,0), VDR(2,7,0)])

# Scalar projection:  $2/3 + 3/5 + 2/7 = 70/105 + 63/105 + 30/105 = 163/105$ 

#  $163 = 1 \cdot 105 + 58 \dots$  but CRT operates on integers mod product

# The structural correspondence is: each child carries one residue channel

```

Appendix K: Gaussian Elimination Pivot Growth for Hilbert Matrices

Intermediate value sizes during Gaussian elimination of Hilbert matrices, showing that VDR handles the growth exactly while float loses significance.

Matrix	Max Pivot Numerator Digits	Max Pivot Denominator Digits	Determinant Numerator	Determinant Denominator	Float det Error
H ₃	3	3	1	2160	$\sim 10^{-16}$
H ₄	5	7	1	6048000	$\sim 10^{-13}$
H ₅	8	11	1	$\sim 2.7 \times 10^{10}$	$\sim 10^{-9}$
H ₆	11	16	1	$\sim 1.9 \times 10^{15}$	$\sim 10^{-4}$
H ₈	18	27	1	$\sim 3.6 \times 10^{26}$	$\sim 10^4$ (wrong sign possible)
H ₁₀	26	40	1	$\sim 4.6 \times 10^{41}$	meaningless
H ₂₀	70	110	1	$\sim 10^{170}$	impossible
H ₃₀	120	185	1	$\sim 10^{400}$	impossible

The Hilbert matrix determinant numerator is always 1. The denominator grows rapidly but remains an exact integer in VDR. Float loses all significance by H₈ and cannot compute H₁₀ determinant at all. VDR computes H₃₀ without difficulty — 120-digit numerators and 185-digit denominators are routine for exact integer arithmetic.

Appendix L: Quaternion Rotation Algebra in VDR

Quaternion operations for 3D rotation with VDR types, supporting slerp and RoPE computations from Sections 6 and 7 of the main paper.

Operation	Formula	VDR Ops	Depth Impact	Closure
Quaternion multiply	$(a_1a_2-b_1b_2-c_1c_2-d_1d_2, \dots)$	16 mul + 12 add/sub	+1 per multiply	Closed if all components closed
Conjugate	$(a, -b, -c, -d)$	3 negations	0	Preserves state
Norm ²	$a^2+b^2+c^2+d^2$	4 mul + 3 add	+1	Real VDR
Inverse	$q^*/\ q\ ^2$	16+4 mul + 15 add/sub + 1 div	+1	Active if norm irrational
Rotation of vector v	qvq^*	2 quaternion multiplies	+2	Closed if q has rational components
Slerp(q_0, q_1, t)	$\frac{\sin((1-t)\theta)}{\sin\theta} \cdot q_0 + \frac{\sin(t\theta)}{\sin\theta} \cdot q_1$	2 sin + 1 arccos + 8 mul + 4 add	fn remainder depth	Requires freeze for Q335
RoPE at position p, dim d	$\cos(p\theta d), \sin(p\theta d)$ applied to 2D pair	1 sin + 1 cos per dim pair	fn remainder depth	Q335 after freeze

Appendix M: Conservation Law Verification Table

Physical conservation laws that VDR verifies by exact equality rather than residual tolerance.

Conservation Law	Physical System	Float Verification	VDR Verification	Significance
Probability = 1	Quantum state $\langle \psi$	$\psi \rangle$	$1.0 \pm \sim 10^{-15}$	1 exactly
Unitarity $U^\dagger U = I$	Time evolution	$I \pm \sim 10^{-15}$	I exactly	Long evolution chains drift in float
Symplecticity $\det(M) = 1$	Paraxial optics	$1.0 \pm \sim 10^{-12}$ after 1000 elements	1 exactly	Resonator stability boundary exact
Energy conservation	Hamiltonian evolution	$E \pm \sim 10^{-12}$ per step	E exactly per step	Discrete calculus, not continuous

Conservation Law	Physical System	Float Verification	VDR Verification	Significance
Parseval energy	DFT/IDFT	Equal $\pm \sim 10^{-14}$	Equal exactly	Filter design verification
Equilibrium $Ku = F$	Structural statics	Residual $\sim 10^{-10}$	Residual = 0	No iterative refinement needed
Group closure	Crystallographic symmetry	Match within tolerance	Structural equality	No tolerance selection
Orbit closure	Kepler 2-body	Position error $\sim 10^{-12}$	Position error = 0	Multi-orbit propagation exact
Partition function ratio	Thermodynamic identity	Ratio $\pm \sim 10^{-14}$	Ratio exact	Free energy differences exact
Coordinate roundtrip	Helmert transform	~ 1 nm residual	0 residual	Datum chain composition exact

Appendix N: Domain Suitability for Physical Computation

Extended suitability matrix specific to the physical domains in this paper.

Domain	Exact	Practical Size	VDR Advantage	Primary Mechanism
QED coefficients	yes	any loop order with known analytical form	eliminates accumulation in multi-term sums	Q335 add + multiply
Quantum 2×2	yes	unlimited operations	exact unitarity, probability	Complex pairs
Quantum $n \times n$	yes	$n \leq 50$ with Gaussian	exact eigenvalues (ratio-nal/complex), exact evolution	Gaussian + complex
DFT/FFT	yes	$N \leq 2^{10}$ comfortably	exact roundtrip, Parseval, no butterfly rounding	Q335 twiddle + nesting
IIR filters	yes	unlimited steps	drift-free; rational powers collapse	Q335 multiply chain

Domain	Exact	Practical Size	VDR Advantage	Primary Mechanism
Transfer functions	yes	any order	exact frequency response	Complex Horner
State-space	yes	unlimited steps, $n \leq 50$ states	zero drift, exact rank	Gaussian + matrix multiply
Kepler orbits	yes	unlimited orbits	zero closure error	Functional remainder Newton
Structural statics	yes	$n \leq 50$ DOF	exact equilibrium, no refinement	Gaussian solve
Partition functions	yes	2^n states, $n \leq 20$	exact Z, F, S	Functional remainder exp/ln
Crystallography	yes	any symmetry group	exact group closure	Rational/Q335 matrix multiply
Geodesy	yes	any transform chain	exact roundtrip	Rational matrix multiply
Paraxial optics	yes	unlimited elements	exact symplecticity	2×2 matrix power
Resonator stability	yes	any cavity	exact stability boundary	Rational comparison

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